

Multi-Token Prediction on a Consumer GPU: A Quality–Throughput Study of Quantized Gemma 4

Dense vs. MoE, Draft-Depth Sweeps, and Greedy-Output Equivalence on an RTX 5070 Ti

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Abstract

Multi-token prediction (MTP) can accelerate autoregressive inference by verifying draft tokens in parallel. Although intended to preserve the target distribution, exact greedy behavior in a finite-precision quantized runtime remains empirical. We evaluated QAT Q4_0 Gemma 4 12B instruction-tuned (dense) and 26B-A4B instruction-tuned (mixture-of-experts) on one NVIDIA RTX 5070 Ti using a pinned CUDA `llama.cpp` build, sweeping MTP off and maximum draft depths $N=\{1,2,3,4,6,8,12,16\}$ over nine fixed performance prompts and a paired 200-item objective suite. $N=6$ produced the highest measured aggregate decode throughput for both models: 226.39 versus 89.20 tokens/s for 12B (2.54x) and 322.53 versus 169.58 tokens/s for 26B-A4B (1.90x). At $N=6$, however, only 112/200 and 118/200 outputs, respectively, were byte-identical to ordinary decoding. Objective macro-score changes were -0.83 percentage points for both models, with paired stratified-bootstrap 95% confidence intervals of [-4.17,+2.71] for 12B and [-2.92,+0.83] for 26B-A4B; both cleared the prospectively frozen -5-point non-inferiority margin as exploratory pilot evidence. Two post-hoc analyses further showed that divergence was associated with longer ordinary-decoding responses and that this negative association persisted within GSM8K, IFEval, and MBPP, but non-randomized length and residual item/response confounding preclude a causal per-token interpretation. They also showed a 12B performance plateau: $N=\{4,6,8,12\}$ lay within 2.6% of the observed maximum, although only $N=6$ received the 200-item quality evaluation. Thus $N=6$ is the evidence-backed high-throughput default in this tested stack, not a proven unique optimum; the findings do not generalize to BF16, other GPUs, runtimes, workloads, or concurrent serving.

1. Introduction

Autoregressive language-model decoding performs a sequential model evaluation for each output token. Speculative decoding reduces this serial cost by asking a faster draft mechanism to propose multiple future tokens, then verifying those candidates with the target model in parallel. The original algorithms are designed to preserve the target distribution (Leviathan et al. 2022; Chen et al. 2023). Multi-token prediction provides a compact, model-linked drafter and has been reported to accelerate inference (Gloeckle et al. 2024). Gemma 4 distributes MTP drafters for its model family and describes substantial decoding speedups without quality change (Google DeepMind 2026).

The deployment question is narrower and more practical: what happens when a particular quantized target,

quantized drafter, backend, and runtime are used on a consumer GPU? “Lossless” can refer to an algorithmic sampling property, but operators may also expect byte-identical greedy text. Quantized matrix operations and different execution paths can change near-tied logits. An upstream `llama.cpp` issue reports such divergence for quantized targets, including Gemma 4 MTP ([llama.cpp issue reporter and contributors 2026](#)). This motivates separate tests of speed, exact behavioral equivalence, and objective task quality.

This study makes five scoped contributions:

1. an operational separation of algorithmic losslessness, deterministic behavioral equivalence, and empirical task-quality equivalence;
2. a depth sweep for two QAT Q4_0 Gemma 4 models on a 16-GB-class RTX 5070 Ti;
3. sequence-level and token-level analysis of greedy-output divergence, including a clearly labeled post-review analysis of its association with response length;
4. paired objective scoring on four task families with a frozen non-inferiority rule; and
5. a reproducible artifact package containing raw outputs, logs, scripts, manifests, tables, and figures.

The aim is not to maximize a headline tokens/s number. It is to identify the quality-constrained operating point for this machine and to state precisely which forms of equivalence the evidence does and does not support.

2. Background

2.1 Three meanings of “lossless”

We distinguish three claims.

Algorithmic losslessness is a property of a speculative sampling method: under its assumptions, the output distribution equals the target model’s distribution. **Deterministic behavioral equivalence** asks whether the exact same greedy token sequence is obtained from two concrete runtime paths. **Empirical task-quality equivalence** asks whether potentially different strings retain the same measured usefulness on scored tasks. Neither of the latter two follows automatically from the first in finite-precision software.

2.2 MTP and verification

MTP trains or supplies heads that predict several future tokens. During inference a smaller drafter proposes a sequence, while the target verifies the proposals in a batch. Speed depends on the cost of drafting and verification, the number of accepted tokens, memory behavior, and prompt/output shape. Larger N can expose more parallel work but can also waste more draft compute after an early rejection. Therefore, the best N is an empirical workload and implementation property, not a universal model constant.

2.3 Quantized local inference

Both targets and both MTP drafters in this study use Q4_0 GGUF artifacts. The target and drafter therefore do not reproduce the source higher-precision arithmetic exactly. We do not attempt to isolate whether observed divergence comes from quantization, kernel order, KV/state management, verification logic, or their interaction. That would require a higher-precision replication and implementation-level tracing.

3. Related work

Leviathan et al. introduced speculative decoding with a proof-oriented account of preserving the target distribution and reported 2–3x acceleration on their evaluated system ([Leviathan et al. 2022](#)). Chen et al. independently described speculative sampling using modified rejection sampling and reported 2–2.5x decoding speedup on a distributed 70B setting ([Chen et al. 2023](#)). These results establish the algorithmic motivation but not exact equivalence for this quantized runtime.

Gloeckle et al. studied multi-token prediction as a training objective and as an inference accelerator (Gloeckle et al. 2024). The Gemma 4 technical report describes the broader dense and MoE model family (Gemma Team 2026), while the official MTP model card describes a smaller draft model whose proposals are verified by the target (Google DeepMind 2026). The local runtime is `llama.cpp`, whose speculative decoding interface explicitly supports draft-MTP and tunable maximum draft depth (ggml-org contributors 2026).

The objective suite uses GSM8K mathematics (Cobbe et al. 2021), MMLU-Pro knowledge/reasoning (Wang et al. 2024), verifiable instruction-following from IFEval (Zhou et al. 2023), and executable MBPP programming tasks (Austin et al. 2021). The study does not claim to reproduce each benchmark’s canonical leaderboard protocol; it uses fixed paired subsets and documents the scoring adaptations.

4. Research questions and hypotheses

- **RQ1 (acceleration):** Does a tested MTP condition improve paired median decode throughput by at least 20% without an increased request-failure rate?
- **RQ2 (depth):** Which tested N maximizes throughput, and does a lower-depth, lower-variance point qualify as conservative?
- **RQ3 (behavioral equivalence):** Are 100% of outputs byte-identical and target-token-identical to a reproducible ordinary-decoding baseline?
- **RQ4 (task quality):** Is the lower limit of a paired bootstrap 95% CI for macro score change greater than a frozen -5 percentage-point margin?
- **RQ5 (model association):** How do the two model/runtime combinations differ, without treating the dense/MoE distinction as a controlled causal intervention?

One verified divergence with a stable ordinary baseline falsifies exact equivalence. Failure of the non-inferiority CI to clear -5 points yields an inconclusive result rather than proof of harm.

5. Methods

5.1 Hardware, software, and models

Experiments ran on Ubuntu 24.04.1 LTS (kernel 6.17.0-40), an Intel Core i7-11700 (8 physical cores, 16 logical CPUs), 125 GiB system memory, and an NVIDIA GeForce RTX 5070 Ti. `llama.cpp` reported 15,841 MiB GPU memory. The pinned runtime was commit `7ba604f1cb61cd14898138e9abc0b4ff2601f180`, built in Release mode with GCC 13.3, CUDA architecture SM120, CUDA and Flash Attention enabled. The server binary SHA-256 was `1590bb2b1f9f704ed204fec890b2bb8cfac93bc2ca08dc3f70ef8053a0824a`.

The 12B target was 6,975,877,728 bytes and the 26B-A4B target was 14,439,361,440 bytes. Their SHA-256 digests, along with those of the 254-MB and 252-MB MTP drafters, are in `configs/environment.json`. Startup logs verified full requested layer placement: 49/49 main and 5/5 draft layers for 12B; 31/31 main and 5/5 draft layers for 26B-A4B. Ollama was not used for inference; the independent `llama-server` binary opened the stored model blobs directly. The allocator reported 6,637.69 MiB of main-model device buffer plus 226.90 MiB of draft buffer for 12B, and 13,755.42 MiB plus 225.21 MiB for 26B-A4B. These are component-buffer reports rather than independent peak process-VRAM measurements.

NVML telemetry was unavailable because the loaded kernel module (595.71.05) and user-space library (595.84) did not match. Consequently, GPU power, temperature, clock, utilization, and independent peak-VRAM series could not be recorded. CUDA execution remained functional, and placement was validated from server logs. This prevents a complete thermal-confound assessment.

5.2 Fixed runtime settings

All main runs used context 8192, concurrency one, temperature 0, top-k 1, fixed per-prompt seeds, Flash Attention, F16 target and draft KV, batch and micro-batch 512, eight CPU threads, and full requested GPU

offload. Model, runtime, and protocol hashes were frozen before new outputs. Conditions were started in deterministic randomized order, and one warm-up request per server start was excluded.

5.3 Performance sweep

For each model, we tested MTP off and N in $\{1,2,3,4,6,8,12,16\}$, with p-min=0. Nine distinct fixed prompts covered three input-length targets (approximately 256, 1,024, and 4,000 tokens; three prompts each). Each request forced 256 output tokens by ignoring EOS. The primary performance endpoint was the median paired decode-rate ratio to OFF. Aggregate tokens divided by aggregate decode time, prompt throughput, wall latency, time to first streamed content, acceptance, and run-level variation were secondary measures. No valid run was removed as an outlier.

5.4 Condition selection

The maximum-throughput point was the tested N with the highest aggregate decode rate. A pre-specified conservative point required the smallest N within 95% of that maximum and a request-level coefficient of variation (CV) at most 10%. Neither model produced an eligible condition. Before inspecting quality outputs, we therefore retained N=6 plus an explicitly exploratory low-variance fallback: N=3 for 12B and N=4 for 26B-A4B.

5.5 Objective quality and equivalence

A deterministic 200-item sample contained 60 GSM8K test items, 60 MMLU-Pro test items, 40 IFEval items, and 40 MBPP items. Each model ran ordinary decoding twice (`off_a`, `off_b`) and both retained MTP conditions, giving 1,600 completed quality requests. GSM8K used normalized exact numeric answer; MMLU-Pro used an exact final option label constrained to the grammar `Final answer: [A-J]`; IFEval used the official Google Research instruction-checking implementation; MBPP ran the reference tests in a network- and filesystem-isolated Bubblewrap sandbox with CPU, memory, and wall limits. Dangerous imports and calls were rejected by an AST pre-check. Scorer fixtures included known pass and fail cases and all passed.

The MMLU-Pro measurement is constrained-choice accuracy, not a reproduction of the benchmark’s chain-of-thought protocol. Output budgets were 512 tokens for GSM8K, 128 for constrained MMLU-Pro, 768 for IFEval, and 512 for MBPP. Smoke testing exposed chat-channel and truncation problems; all corrections were versioned before the full analyzed run and are disclosed in `analysis/DEVIATIONS.md`.

Exact equivalence metrics were byte equality, target-token equality, common prefix and first divergent token, normalized character Levenshtein similarity, and token-length difference. Ordinary-decoding repeat agreement defined a stable baseline subset. A divergent continuation was counted once per sequence.

5.6 Statistical analysis

The primary quality score was the unweighted mean of the four family accuracies. For each retained MTP condition, we computed the paired macro difference and a task-stratified paired bootstrap 95% confidence interval with 10,000 resamples and seed 20260809. The frozen non-inferiority margin was -5 percentage points. Within each model, family-level paired changes used exact McNemar tests with Holm correction. Because N=200 was resource-bounded rather than prospectively powered for small effects, passing the margin is treated as exploratory non-inferiority evidence, not definitive equivalence.

5.7 Negative controls and contamination diagnostics

The principal negative control was an independent OFF repeat. We also rotated the option order for 20 MMLU-Pro items and compared the implied semantic answer with the original order under OFF and N=6. Prompt duplicate and shared n-gram diagnostics were run. Public benchmark exposure in the model’s training data cannot be ruled out, so absolute capability and state-of-the-art claims are prohibited.

5.8 Post-review exploratory analyses

The first external review added two analyses; neither is part of the frozen confirmatory protocol. First, for each retained MTP condition, we grouped the 200 quality pairs by the natural OFF-A output length (1–8, 9–32, 33–64, 65–128, or at least 129 target-token IDs) and calculated exact byte-match rates. We also calculated Spearman association and a descriptive Kaplan–Meier-like no-divergence curve in which exact pairs were censored at their natural ending. Natural endings can be informative censoring, and task family and response form are strongly confounded with length; the analysis therefore cannot identify an independent or constant per-token divergence probability.

Second, we resampled the nine matched performance prompts 100,000 times with replacement (seed 20260809), recomputed aggregate decode rate at every tested positive N, and recorded which N was fastest. We also bootstrapped the paired median speed ratio between the observed best and runner-up depths. This quantifies workload-sampling uncertainty only; it does not include fresh repeated-run, thermal, or system-state variation.

Following a second external review, we partially addressed task-family confounding without collecting new outputs. Within each model, retained condition, and family, we calculated Spearman correlation between OFF-A length and exact equality, with 20,000 pair-bootstrap resamples for 95% intervals. Benjamini–Hochberg correction covered all 12 identifiable family comparisons; MMLU-Pro was non-identifiable because every OFF output had five token IDs. We also converted length to a within-family fractional rank, centered both that rank and equality within family, and correlated the pooled residuals across GSM8K, IFEval, and MBPP (N=140). This task-stratified statistic used 20,000 stratified bootstrap resamples, 100,000 within-family permutations, and BH correction across four retained model/condition comparisons. It removes between-family level differences but does not randomize length or eliminate within-family item complexity, output-form, stopping, and truncation confounding.

Finally, we summarized the 12B conditions at least 95% as fast as the observed grid maximum, reusing the prospectively specified speed threshold. This is a descriptive performance band, not an equivalence test. Conditions without the 200-item objective run cannot inherit N=6’s quality result.

6. Performance results

All 162 planned main performance requests were valid (81 per model). Table 1 shows the selected rows; complete results appear in `tables/performance_by_depth.md`.

Model	N	Aggregate decode tok/s	Paired median speedup (95% bootstrap CI)	CV	Acceptance	Median wall ms	Median TTFC ms
12B	OFF	89.20	–	0.019	–	3348	526
12B	3	203.91	2.261	0.070	0.717	1800	454
			[2.142, 2.468]				
12B	6	226.39	2.497	0.155	0.577	1985	518
			[2.266 , 3.054]				
26B-A4B	OFF	169.58	–	0.027	–	1847	369
26B-A4B	4	300.92	1.789	0.098	0.730	1300	367
			[1.606, 1.968]				

Model	N	Aggregate decode tok/s	Paired median speedup (95% bootstrap CI)	CV	Acceptance	Median wall ms	Median TTFC ms
26B-A4B	6	322.53	1.860 [1.701, 2.202]	0.122	0.706	1227	370

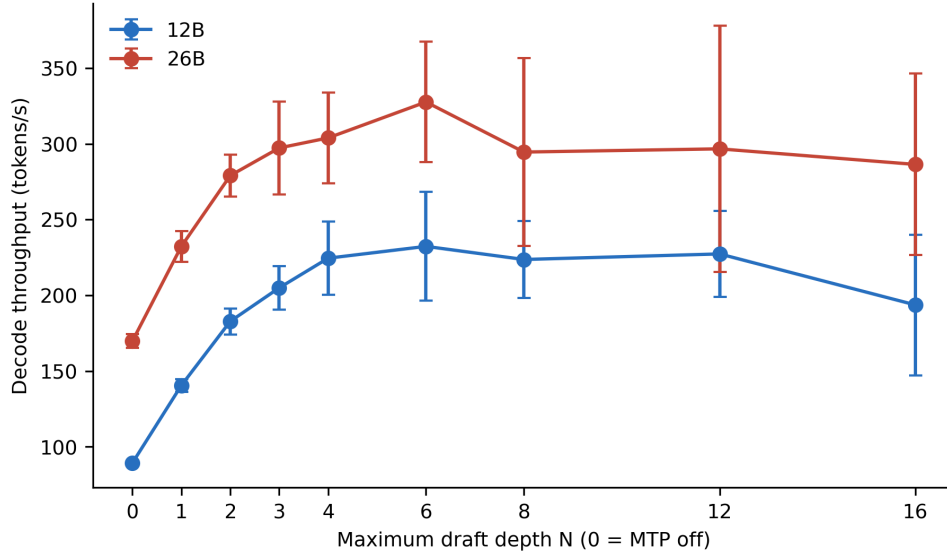


Figure 1: Draft depth versus decode throughput. Error bars are run-level standard deviations.

N=6 maximized aggregate throughput for both models. Relative to OFF, aggregate ratios were 2.54x for 12B and 1.90x for 26B-A4B. Both also passed the pre-set 20% practical-acceleration threshold. Throughput fell or plateaued beyond N=6, while acceptance generally declined. For 12B, N=6 proposed 3,077 draft tokens and accepted 1,775 (57.7%); for 26B-A4B it proposed 2,626 and accepted 1,853 (70.6%). N=16 did not maximize either workload.

TTFC changed little relative to baseline at the maximum point, whereas full-request latency fell substantially because 256 output tokens completed faster. The higher CV at N=6 explains why the frozen conservative rule returned no winner. Interpreting differences among nearby high-N points requires caution because variation was not negligible. In the post-review prompt bootstrap, 12B N=6 was fastest in 51.3% of resamples, versus 29.4% for N=12, 13.5% for N=8, and 5.8% for N=4. The paired median speed ratio of N=6 to N=12 was 1.066, with a 95% prompt-bootstrap interval of [0.803, 1.202]. For 26B-A4B, N=6 was fastest in 93.9% of resamples; its paired median ratio to runner-up N=4 was 1.019 [0.972, 1.310]. Thus N=6 is the observed grid maximum for both models, but its advantage over neighboring depths is not resolved as a population ordering, especially for 12B.

The observed 12B performance band defined by the frozen 95%-of-maximum speed threshold contained four, not three, depths:

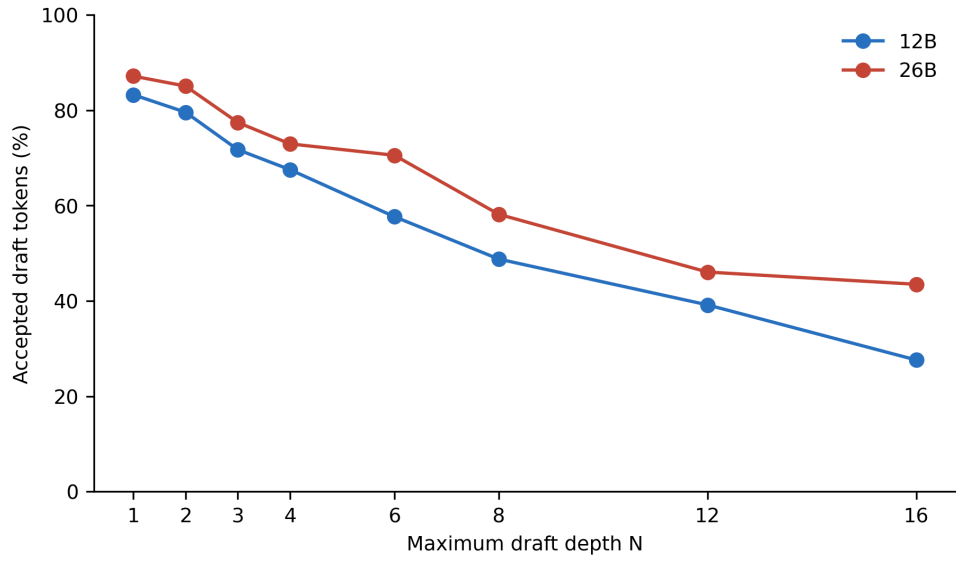


Figure 2: Draft depth versus accepted share of proposed tokens.

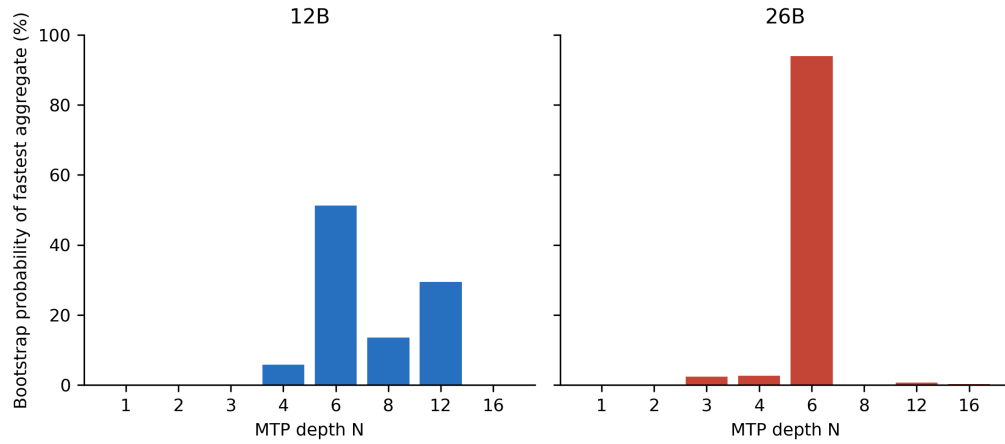


Figure 3: Post-review prompt-bootstrap probability that each positive depth is the fastest aggregate condition. This reflects the nine-prompt workload sample, not repeated-system variability.

12B N	Aggregate tok/s	Share of observed max	CV	200-item objective quality
4	221.69	97.92%	10.80%	Not evaluated
6	226.39	100.00%	15.49%	Evaluated; margin cleared
8	220.67	97.47%	11.31%	Not evaluated
12	223.30	98.63%	12.44%	Not evaluated

This supports a 12B **performance-tuning plateau** at $N=\{4,6,8,12\}$. It does not support treating all four as quality-equivalent or interchangeable: $N=4$, $N=8$, and $N=12$ did not receive the paired 200-item objective evaluation, and none passed the full conservative rule because every CV exceeded 10%. $N=6$ therefore remains the quality-evaluated high-throughput default; $N=3$ remains the quality-evaluated lower-variance alternative.

7. Deterministic-equivalence results

Ordinary decoding reproduced exactly on 200/200 quality items for 12B and 199/200 for 26B-A4B. The one unstable 26B item is retained in the overall data but excluded from stable-subset claims.

At $N=6$, 12B matched OFF byte-for-byte on 112/200 outputs (56.0%); 26B-A4B matched on 118/200 (59.0%), or 117/199 (58.8%) among stable baselines. Target token equality had the same rates. Median first divergence among divergent sequences was token 42 for 12B and 37.5 for 26B-A4B. Mean normalized character similarity remained 0.851 and 0.869, respectively, and median output-length difference was zero. The lower-variance conditions did not restore exactness: 12B $N=3$ matched 55.0%, and 26B-A4B $N=4$ matched 59.0%.

On the nine forced-length performance prompts, 12B $N=6$ matched none of the OFF outputs; 26B-A4B $N=6$ matched one of nine. The independent stable baseline makes the quality-suite divergences sufficient to reject exact behavioral equivalence for this stack.

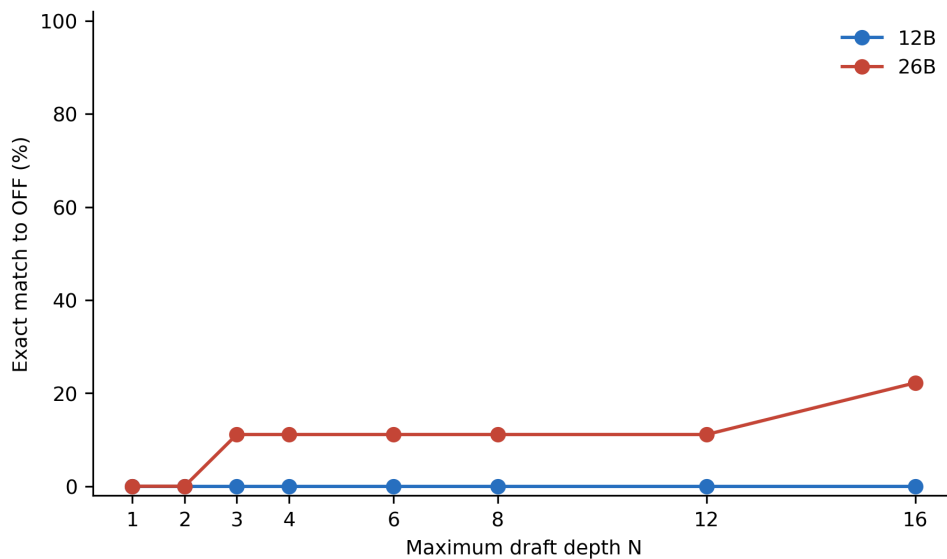


Figure 4: Exact output match on the forced-length performance prompts.

The divergence result is not a quality result. Autoregressive branching can produce different but equally valid completions after one token changes. The sequence-level metrics are therefore reported separately from correctness.

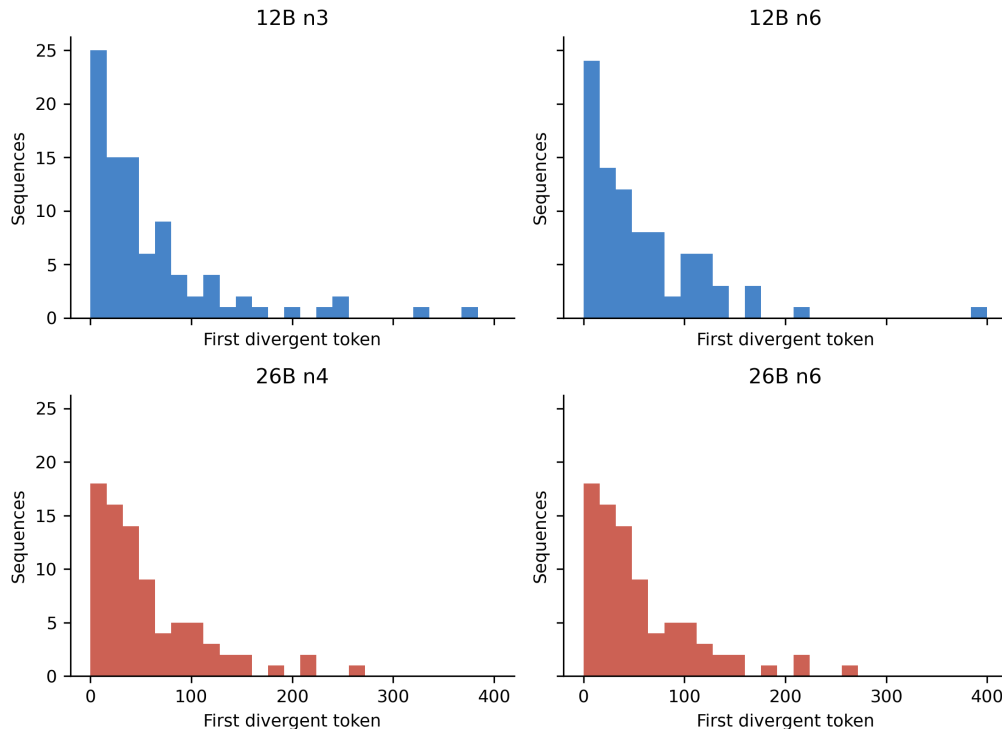


Figure 5: Distribution of the first divergent target-token position among divergent sequences. Exact matches are excluded from the histogram and reported separately.

The post-review length analysis revealed a strong descriptive gradient. At $N=6$, 12B matched 60/61 OFF outputs of 1–8 tokens and all 12 outputs of 9–32 tokens, but only 24/98 outputs of at least 129 tokens. For 26B-A4B, the corresponding counts were 61/61, 5/6, and 22/88. Spearman correlations between OFF length and exact equality were -0.703 for both $N=6$ conditions (unadjusted exploratory p-values below 10^{-30}). Exact pairs had median OFF length five tokens, whereas divergent pairs had medians 235.5 (12B) and 220.5 (26B-A4B). The forced 256-token performance results are consistent with more opportunity to diverge over a long continuation. However, short outputs here are dominated by constrained-answer tasks while long outputs come from generative tasks. The data therefore support an operational length association, not the causal claim that every token carries an independent fixed divergence probability.

The second-review family-stratified analysis showed that the gradient was not solely a between-family composition artifact. At $N=6$, within-family Spearman correlations were -0.329, -0.566, and -0.701 for 12B and -0.321, -0.513, and -0.643 for 26B-A4B in GSM8K, IFEval, and MBPP, respectively. All six 95% pair-bootstrap intervals excluded zero, and all six exploratory BH-adjusted p-values were at most 0.0135. MMLU-Pro could not identify a length association because all 60 OFF outputs had length five.

After centering length rank and equality within the three variable-length families, the pooled $N=6$ correlations were -0.490 [95% stratified-bootstrap CI -0.610, -0.343] for 12B and -0.450 [-0.581, -0.297] for 26B-A4B. Both within-family permutation tests reached the minimum attainable two-sided Monte Carlo p-value of 0.000010 with 100,000 permutations, unchanged by BH correction across the four retained comparisons.

8. Objective task-quality results

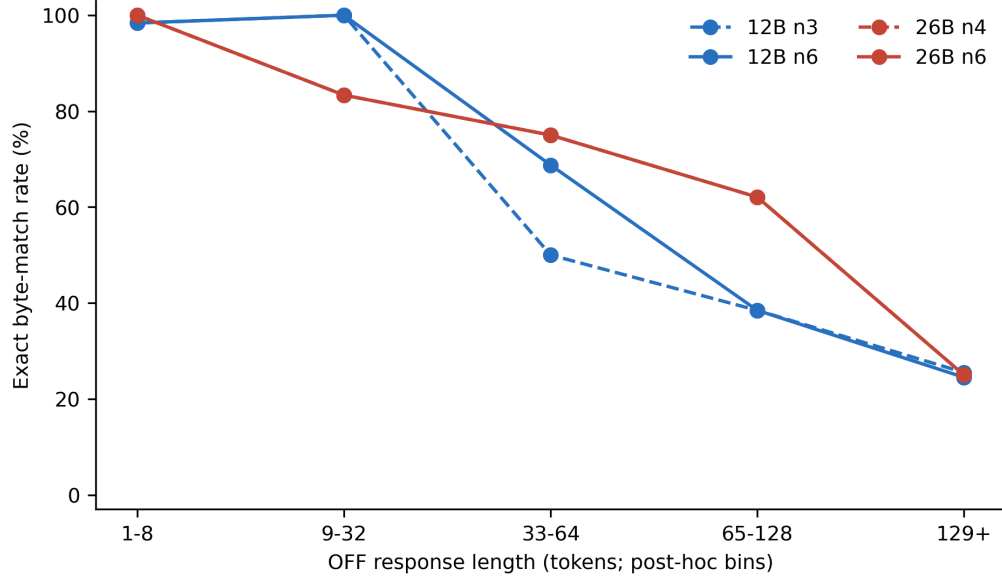


Figure 6: Post-review exact-match rates by natural OFF response-length bin. Length is confounded with task family and output form; bin counts are reported in the artifact.

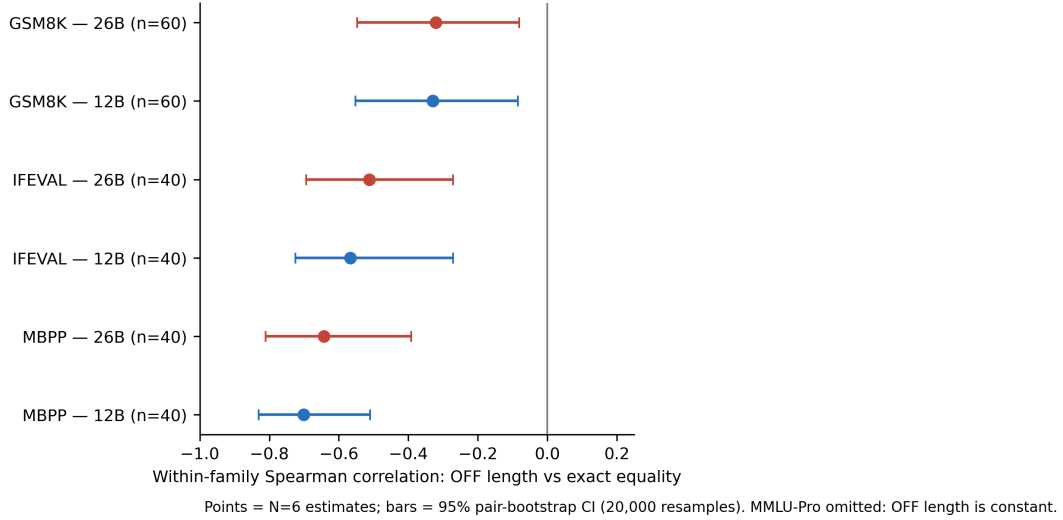


Figure 7: Within-family association between natural OFF response length and exact equality at N=6. Points are Spearman correlations; bars are 95% pair-bootstrap intervals from 20,000 resamples. MMLU-Pro is omitted because OFF length is constant.

Model	Condition	GSM8K	MMLU-Pro	IFEval	MBPP	Macro	Macro change (pp)	95% CI (pp)
12B	OFF	93.33	56.67	85.00	85.00	80.00	—	—
12B	N=3	93.33	55.00	87.50	82.50	79.58	-0.42	[-3.33, +2.50]
12B	N=6	91.67	55.00	90.00	80.00	79.17	-0.83	[-4.17, +2.71]
26B-A4B	OFF	88.33	61.67	87.50	85.00	80.63	—	—
26B-A4B	N=4	90.00	61.67	85.00	82.50	79.79	-0.83	[-2.92, +0.83]
26B-A4B	N=6	90.00	61.67	85.00	82.50	79.79	-0.83	[-2.92, +0.83]

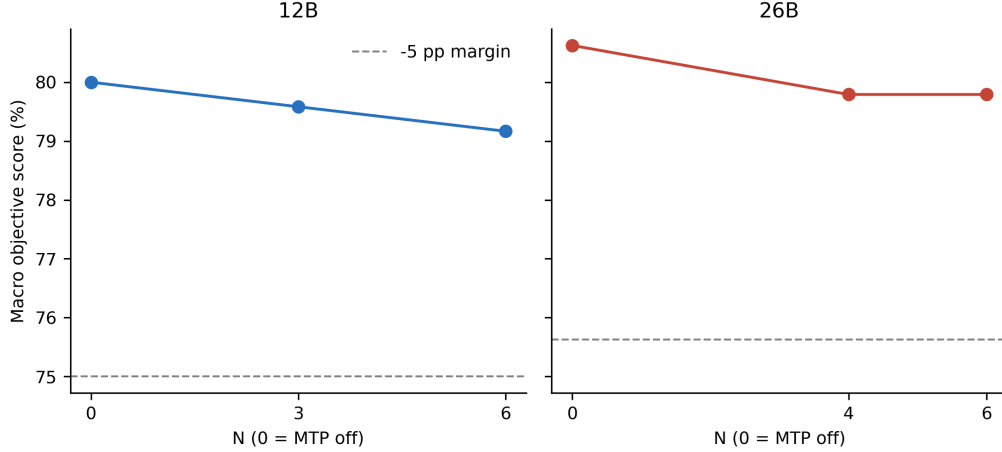


Figure 8: Objective macro score at the retained conditions. The dashed line is five percentage points below baseline.

All four MTP comparisons cleared the frozen -5-point lower-bound rule. At N=6, 12B had six regressions and four improvements across the 200 paired items; 26B-A4B had two regressions and one improvement. The macro score differs from the raw 200-item micro proportion because task families are equally weighted. Every family-level exact McNemar p-value became 1.0 after within-model Holm correction. Thus, this pilot found no statistically established family-level effect, while its uncertainty permits small harms or gains.

There were length-capped responses even with corrected budgets: six for each 12B OFF repeat, nine at 12B N=6, seven at N=3, and four in each 26B-A4B condition. They were retained and scored under the fixed rules. Parse, execution, timeout, and truncation outcomes were not silently dropped.

9. Dense and MoE model-level association

The 12B dense model realized the larger relative acceleration at N=6 (2.50x paired median; 2.54x aggregate) despite a lower draft acceptance rate (57.7%). The 26B-A4B model realized a smaller relative gain (1.86x paired median; 1.90x aggregate) with 70.6% acceptance. The absolute OFF decode rate was already higher for 26B-A4B, consistent with its sparse active computation, but the experiment did not hold model size, training, active parameters, or kernels constant. One plausible systems interpretation is that the already-fast MoE baseline leaves less relative serial work for batched verification to remove, despite high draft acceptance.

This mechanism was not isolated. These are model/runtime associations, not proof that MoE architecture causes a smaller MTP speedup.

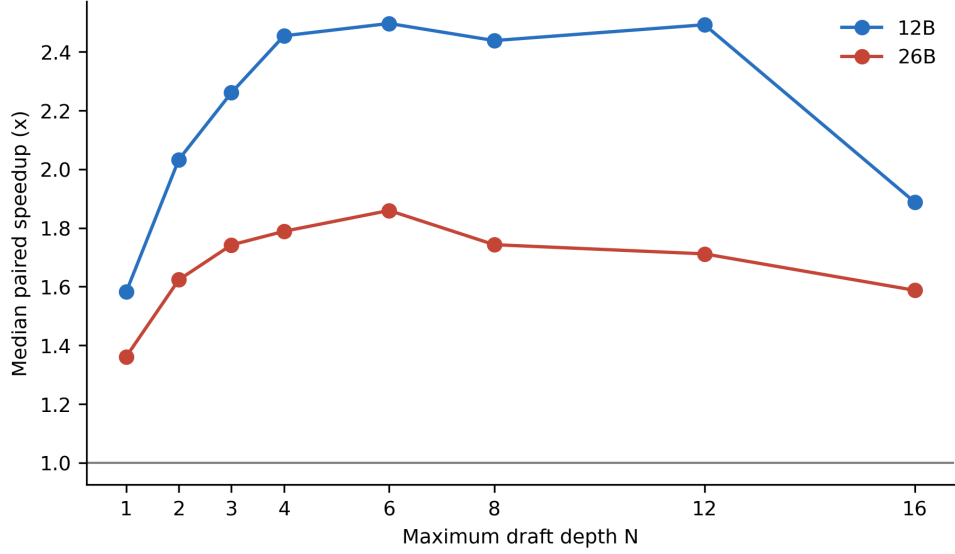


Figure 9: Paired speedup by draft depth for the two tested model/runtime combinations.

10. Quality–throughput Pareto analysis and operating points

On the measured objective suite, $N=6$ is the fastest tested condition whose paired CI clears the frozen non-inferiority margin for both models. It is the **quality-constrained maximum-throughput point**, not a universal optimum. For 12B, $N=\{4,6,8,12\}$ form an observed performance plateau under the 95% threshold, but only $N=6$ is quality constrained by the 200-item evidence.

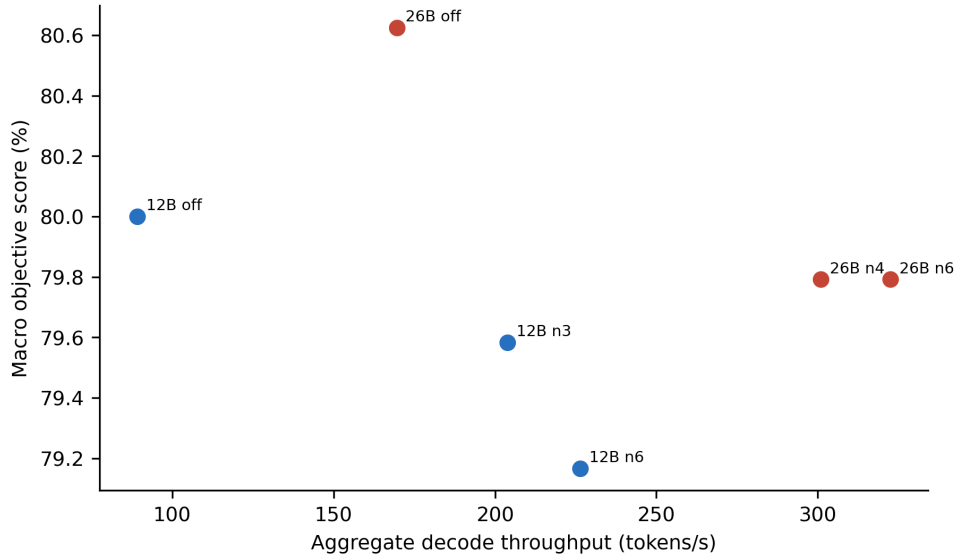


Figure 10: Measured quality–throughput points for OFF and retained MTP conditions.

For lower run-to-run variation, 12B $N=3$ (203.91 tokens/s, CV 7.0%) and 26B-A4B $N=4$ (300.92 tokens/s, CV 9.8%) are reasonable exploratory operational alternatives. Neither is formally the **conservative deployment**

point under the frozen rule because each fell below 95% of the N=6 aggregate maximum. Moreover, neither reduced output divergence. The study therefore reports “no eligible strict conservative point” rather than changing the rule after seeing the data.

11. Exploratory p-min sweep

The optional phase was triggered because N at least 6 reached 90% of each model’s maximum. We evaluated 144 additional valid requests. For 12B, the best exploratory combination was N=12, p-min=0.75 at 230.25 tokens/s, only 1.70% above the N=6, p-min=0 reference and with CV 17.2%. For 26B-A4B, N=6, p-min=0.50 reached 330.50 tokens/s, 2.47% above reference, with CV 10.3%.

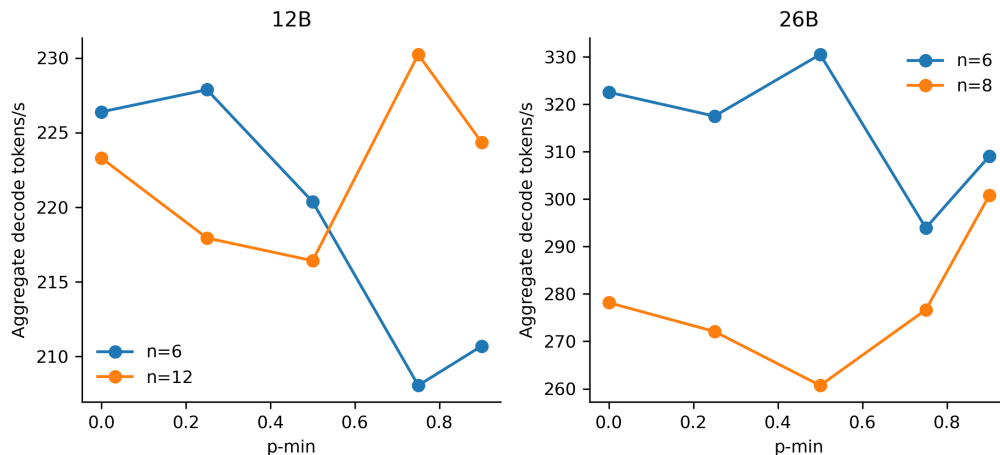


Figure 11: Exploratory p-min results.

Because these percentage gains are small relative to observed variation and the p-min conditions did not receive the 200-item quality evaluation, neither replaces the primary N=6 recommendation. The 26B-A4B p-min=0.50 result is a candidate for a preregistered replication.

12. Negative controls and contamination limits

The ordinary baseline repeat was exact for all 12B items and all but one 26B item, showing that widespread OFF instability does not explain the MTP divergence. In the 20-item option-order perturbation, 12B semantic-option stability versus original order was 65% under OFF and 60% under N=6; the perturbed MTP text matched perturbed OFF in 19/20 cases. For 26B-A4B, semantic stability was 65% for both and MTP matched OFF in 20/20 cases. These small controls show prompt-order sensitivity but no distinctive MTP failure signal.

No exact prompt duplicates were found. A conservative normalized 13-gram scan found four shared substantive phrases after wrapper removal. Such diagnostics cannot determine whether public benchmark items occurred in training. The paired intervention comparison is less sensitive to shared contamination than an absolute model-quality claim, but training exposure remains unresolved.

13. Discussion

The primary practical result is that MTP moved both models well beyond the pre-set acceleration threshold while retaining objective scores within the exploratory non-inferiority margin. For single-stream, 256-token continuations on this GPU, N=6 is a defensible evidence-backed default. The depth conclusion is asymmetric: 26B-A4B N=6 was the prompt-bootstrap winner in 93.9% of resamples, whereas 12B has a broad measured

performance plateau at $N=\{4,6,8,12\}$. Operators re-tuning 12B for a different workload can benchmark that band, but $N=4$, $N=8$, and $N=12$ cannot inherit $N=6$'s objective-quality result.

The behavioral result is equally important: exact greedy text cannot be promised for these Q4_0 target/drafter pairs in this runtime. Byte/token match rates near 56–59% on the quality suite are too low to dismiss as isolated noise. Systems that require reproducible hashes, stable diffs, deterministic audit trails, or cache keys derived from generated text should treat MTP as a behavior-changing execution mode, even though task scores were stable here. Within this suite, short constrained outputs were usually identical whereas long generative outputs often diverged. That pattern is compatible with cumulative opportunity for path-dependent numerical differences to alter a continuation. The negative association remained within GSM8K, IFEval, and MBPP, so between-family differences in average length and match rate are not a sufficient explanation. Nevertheless, longer responses within each family can proxy item complexity, response structure, stopping behavior, or truncation exposure. The analysis therefore strengthens a descriptive length association without isolating a causal per-token mechanism.

The quality finding should not be overstated. Passing a -5-point margin in 200 paired items supports a local deployment decision but does not prove identical intelligence, semantic equivalence on every answer, or absence of smaller systematic regressions. The open intervals contain both minor harm and minor benefit. A larger confirmatory suite, preferably with private or recently constructed tasks, is needed for narrower claims.

14. Limitations and threats to validity

External validity. The study covers one RTX 5070 Ti, one OS, one pinned llama.cpp commit, QAT Q4_0 targets and drafters, concurrency one, context 8192, text-only inputs, and one fixed prompt/output distribution. It does not cover BF16, other quantizations, multimodal inputs, long-context saturation, high-concurrency service throughput, other GPUs, or other speculative runtimes.

Measurement validity. NVML failure prevented temperature, power, clock, background-utilization, and independent peak-VRAM logging. Randomized condition order, warm-up exclusion, per-run preservation, and variation reporting reduce but do not eliminate thermal or system-load confounding. Streaming TTFC is time to first content observed by the client, not a kernel-level first-token measurement.

Construct validity. Objective benchmarks cover math, constrained knowledge, verifiable instructions, and Python synthesis. They do not measure open-ended Japanese prose, summarization quality, safety, factual calibration, dialogue, or human preference. An uncalibrated LLM-as-judge phase was deliberately omitted. MMLU-Pro was grammar-constrained and therefore should not be compared directly with canonical chain-of-thought leaderboard scores.

Statistical conclusion validity. Four families and 200 items provide a pilot, not guaranteed power for sub-percentage effects. The margin is practical and frozen, but a -5-point lower bound may still be too permissive for high-stakes tasks. The $N=6$ lower bounds of -4.17 and -2.92 points exclude the frozen -5-point margin but do not exclude smaller systematic losses. A tighter margin requires a larger, use-case-specific suite. Performance confidence intervals use only nine paired prompts per condition. The post-review prompt bootstrap confirms that the 12B ordering among $N=6$, $N=12$, and nearby depths is not secure; it also omits run-to-run and thermal uncertainty.

Internal validity. One 26B ordinary response was not reproducible. Stable subset rates are reported. The condition-selection fallback was specified only after the strict rule returned no eligible point, but before quality outputs were examined; it is explicitly exploratory. Smoke-test corrections to output budgets, reasoning-channel parsing, and constrained MMLU output are fully versioned, but they weaken confirmatory framing.

Mechanism. No higher-precision control or kernel trace was run. The study cannot attribute divergence to quantization alone, nor can it determine whether the behavior is fixed in a later runtime. The response-length analysis was post-review. Task-family stratification removes between-family level differences but not within-family complexity, response-form, stopping, or truncation confounding; its descriptive survival estimate may

also have informative censoring at natural endings. It does not distinguish quantization, kernel ordering, verification batching, or KV/state management.

15. Reproducibility

The private internal archive contains the frozen protocol, environment and model hashes, fixed prompts, source revisions, launch scripts, append-only raw JSONL, complete server logs, deterministic scorer code, processed CSV/JSON, publication figures, and validation reports. The separate public reproducibility archive omits verbatim benchmark items, generated response text, complete logs, local paths, and vendored third-party source. It retains task identifiers, hashes, measurements, processed results, pinned upstream revisions, retrieval code, and reproducibility instructions. The public redistribution audit documents every excluded field and directory. Raw evidence is hashed in the private archive; public artifacts are hashed separately. The original preliminary benchmark remains outside this study root and was not modified. The post-review analysis is regenerated by `scripts/analyze_review_followup.py`; its status and caveats are embedded in the output JSON.

Because source model paths are machine-local and the model artifacts are large, another operator must supply byte-identical files or update paths and treat the result as a replication condition. Re-running against a newer `llama.cpp` commit is also a replication, not a direct reproduction.

16. Ethics and AI-assistance disclosure

No human participants, private user logs, personal data, or human-subject interventions were used. Generated Python was executed only in a constrained local sandbox. Public benchmark licenses and source provenance remain the responsibility of any redistributor of the artifact package.

OpenAI Codex assisted with experimental scripting, execution orchestration, data-processing and research engineering, analysis, visualizations, manuscript language preparation, and dissemination preparation. All quantitative statements in the paper are derived from machine-readable outputs and validation scripts. AI systems are not authors. The human author, Taiko Toeda, is responsible for the research design, measurements accepted for publication, interpretation, citations, claims, attribution, venue compliance, and the final publication decision.

17. Conclusion

For the tested QAT Q4_0 Gemma 4 stack on an RTX 5070 Ti, MTP at N=6 raised aggregate decode throughput from 89.20 to 226.39 tokens/s for 12B and from 169.58 to 322.53 tokens/s for 26B-A4B. The paired 200-item objective suite cleared a frozen -5-point non-inferiority margin, with observed macro changes of -0.83 points for both models. Yet only 56% and 59% of selected MTP outputs were exactly identical to ordinary decoding. The evidence therefore supports a precise deployment statement: MTP is substantially faster and did not show a meaningful objective-quality loss under this pilot, but it is not behaviorally identical in this quantized runtime. N=6 is the measured quality-constrained observed maximum in this grid; its near-depth ranking is not definitive, particularly for 12B. The 12B performance-only plateau is $N=\{4,6,8,12\}$, but only N=6 in that band has the objective-quality evidence; N=3 is the quality-tested lower-variance alternative. For 26B-A4B, N=6 is the more robust depth winner and N=4 is the lower-variance alternative. None is a formal conservative winner. Higher-precision, cross-runtime, and length-balanced replication remain the highest-value next tests.

Appendix A. Exact launch configuration

The common server settings were equivalent to:

```

llama-server \
  --model TARGET.gguf \
  --ctx-size 8192 --parallel 1 \
  --flash-attn on --cache-type-k f16 --cache-type-v f16 \
  --batch-size 512 --ubatch-size 512 --threads 8 \
  --n-gpu-layers 999 --fit off --jinja \
  --reasoning off --reasoning-format deepseek

```

MTP conditions additionally supplied the pinned Q4_0 drafter and:

```

--spec-type draft-mtp \
--model-draft DRAFTER.gguf \
--n-gpu-layers-draft 999 \
--spec-draft-n-max N \
--spec-draft-p-min 0

```

Requests used temperature 0, top-k 1, fixed seeds, and `reasoning_budget_tokens=0`. The performance phase used `ignore_eos=true` and `max_tokens=256`. Exact executable paths and scripted invocations are given in REPRODUCE.md.

Appendix B. Evidence map

- Full performance grid: `processed/performance_summary.csv`
- Request-level performance: `processed/performance_runs.csv`
- Quality family scores: `processed/quality_summary.csv`
- Paired score changes and tests: `processed/paired_quality_changes.csv`
- Sequence equivalence: `processed/equivalence_summary.csv` and `processed/equivalence_pairs.csv`
- Exploratory p-min: `processed/pmin_summary.csv`
- Post-review length association: `processed/length_divergence_by_bin.csv`, `processed/divergence_survival.csv`, and `processed/review_followup_analysis.json`
- Second-review within-family length association: `processed/length_divergence_within_family.csv` and `processed/length_divergence_stratified.csv`
- Post-review depth-rank bootstrap: `processed/depth_rank_uncertainty.csv`
- 12B observed performance plateau: `processed/12b_performance_plateau.csv`
- Negative control: `processed/negative_control_summary.csv`
- Deviations: `analysis/DEVIATIONS.md`
- Environment: `configs/environment.json` and `ENVIRONMENT.md`

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